

Stability Conditions

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Bridgeland stability conditions are a way of giving a further refinement of a t -structure on a triangulated category.

1 Warm-Up

To work up to the definition of a stability condition on a triangulated category, we're going to start with one source of examples, with some working definitions.

1.1 Stability on an Abelian Category

We will soon replace all of these temporary definitions, but they will at least capture special cases.

Definition 1.1. Let $\mathcal{A}$ be an abelian category. The **Grothendieck group** $K(\mathcal{A})$ is the abelian group

$$\mathbb{Z}[\mathcal{A}] \Big/ \left(B = A + C \quad \text{if} \quad 0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0 \right).$$

Example 1.2. Consider $\mathcal{A} = \text{Coh}(\mathbb{P}^1)$.

We first compute $K(\text{Coh}(\mathbb{P}^1))$. Given a sheaf $\mathcal{F}$, we get the Hilbert polynomial

$$p_{\mathcal{F}}(t) := \chi(\mathcal{F}(t)),$$

which will be some polynomial $dt + (d + r)$ for integers $d, r \in \mathbb{Z}$, which we respectively call the *degree* and the *rank*. Since the Hilbert polynomial is additive on exact sequences, i.e., given

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0,$$

we have

$$p_{\mathcal{F}'}(t) - p_{\mathcal{F}}(t) + p_{\mathcal{F}''}(t) = 0,$$

the Hilbert polynomial of the class of a coherent sheaf in the Grothendieck group is well-defined. That is, we have a well-defined abelian group homomorphism

$$K(\text{Coh}(\mathbb{P}^1)) \rightarrow \mathbb{Z}^2$$

which on generators sends $[\mathcal{F}] \mapsto (d, r)$ (or in general we could have instead mapped to $\mathbb{Z}^2 \cong \mathbb{Z}[t]_{<2}$ and just used the Hilbert polynomial itself as the group homomorphism).

Clearly, the group homomorphism is surjective, since $[\mathcal{O}]$ has $d = 0, r = 1$ and $[\mathbb{C}_p]$ has $d = 1, r = 0$, and then every other d, r can be achieved by linear combinations. We want to show that the group homomorphism is injective, so that it is an isomorphism.

To show injectivity, let $E \in K(\text{Coh}(\mathbb{P}^1))$ with $d(E) = r(E) = 0$. We will show $E = 0$.

First, we find a special representation of E . Every coherent sheaf $\mathcal{F}$ on $\mathbb{P}^1$ has a resolution by direct sums of line bundles (see Hilbert's syzygy theorem), so we can write

$$E = \sum_i d_i [\mathcal{O}(a_i)].$$

with finite support. By separating out the positive/negative d_i and putting more redundant terms, we can even write

$$E = \sum_i [\mathcal{O}(a_i)] + \sum_j [\mathcal{O}(b_j)].$$

Now, we have the Koszul complex

$$0 \rightarrow \mathcal{O}(-a-b) \rightarrow \mathcal{O}(-a) \oplus \mathcal{O}(-b) \rightarrow \mathcal{O} \rightarrow 0$$

by taking the associated sheaves of the regular system of parameters x^a, y^b for the irrelevant ideal. This tells us

$$[\mathcal{O}(-a)] + [\mathcal{O}(-b)] = [\mathcal{O}(-a-b)] + [\mathcal{O}].$$

Using this relation, we can rewrite E as

$$\begin{aligned} E &= [\mathcal{O}(\sum a_i)] + r[\mathcal{O}] - [\mathcal{O}(\sum b_j)] - s[\mathcal{O}] \\ &= [\mathcal{O}(\sum a_i)] - [\mathcal{O}(\sum b_j)] + (r-s)[\mathcal{O}]. \end{aligned}$$

Now, the rank 0 assumption tells us $1 - 1 + r - s = 0$ so the last term disappears, and the degree 0 assumption tells us $\sum a_i - \sum b_j = 0$, so the first two terms cancel out. So, indeed $E = 0$.

Example 1.3. On a smooth projective variety X over $\mathbb{C}$, the **chern character** is a ring homomorphism

$$\text{ch} : K(\text{Coh}(X)) \rightarrow H^*(X; \mathbb{Q}),$$

where $H^*(X; \mathbb{Q})$ is often a more well-understood group, so we often don't study all of $K(\text{Coh}(X))$ but instead its image in $H^*(X; \mathbb{Q})$ which will be a finite rank lattice.

Before specifying a *stability condition* on an abelian category $\mathcal{A}$, fix a finite rank lattice Λ and a morphism $K(\mathcal{A}) \xrightarrow{\lambda} \Lambda$.

Definition 1.4. A **stability function** on $\mathcal{A}$ is a group homomorphism

$$Z : \Lambda \rightarrow \mathbb{C}$$

such that

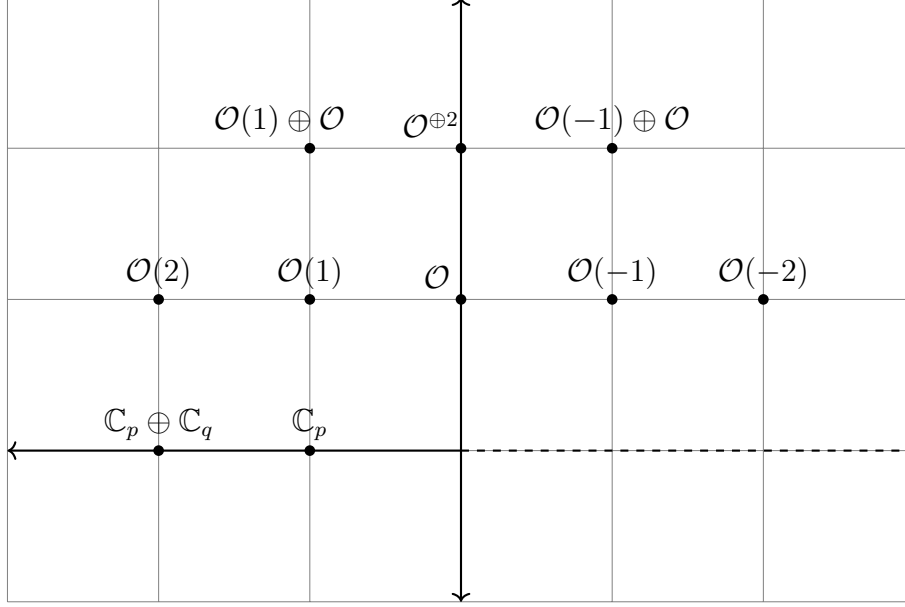
$$\arg_Z(A) := \arg(Z([A])) \in (0, 1]$$

for $A \neq 0$ (where we are normalizing angles by $\arg e^{\pi t} = t$). In other words, a nonzero object of $\mathcal{A}$ is mapped to the upper half plane, removing the positive real axis.

Example 1.5. We can define a stability function on $\text{Coh}(\mathbb{P}^1)$ by

$$\begin{aligned} Z : K(\text{Coh}(\mathbb{P}^1)) &\longrightarrow \mathbb{C} \\ (d, r) &\longmapsto -d + ir, \end{aligned}$$

so graphing it we have something like



where we've plotted where some coherent sheaves land. We will want to think about not just where equivalence classes in the Grothendieck group land, but where a coherent sheaf lands (by sending it to its equivalence class and then using Z). For example, $\mathcal{O}(1)$ and $\mathcal{O} \oplus \mathbb{C}_p$ will both land in the same spot at $-1 + i$.

Definition 1.6. Given a stability function $Z : K(\mathcal{A}) \rightarrow \mathbb{C}$, an object $A \in \mathcal{A}$ is called

- **unstable** if there exists $B \leq A$ such that $\arg(B) > \arg(A)$ (so B “destabilizes” A)
- **semistable** if not unstable.
- **stable** if for all $B < A$, $\arg(B) < \arg(A)$.
- **strictly semistable** if semistable but not stable,

where $\leq$ is an injection, $<$ a nonsurjective injection.

The **semistable objects of phase θ** is denoted

$$\mathcal{A}_\theta := \{A \in \mathcal{A} \mid \arg(A) = \theta \text{ and } A \text{ semistable}\}.$$

Example 1.7. Going back to our $\text{Coh}(\mathbb{P}^1)$ example, the sheaves $\mathcal{O}(a)$ and $\mathbb{C}_p$ are all stable sheaves—there are only injections to them from coherent sheaves of strictly smaller phase.

This also goes back to a more classical notion, called the **Mumford slope**. In our setup, rather than using

$$\arg(\mathcal{F}) = \frac{1}{\pi} \cot^{-1} \left(\frac{-d}{r} \right)$$

one could use $\mu(\mathcal{F}) = \frac{d}{r}$, since $\cot(-x)$ is monotonic. One can still use Mumford slope for more general projective varieties to define notions of stability, but it will turn out to not be what we want—using only degree and rank is fine for curves, but in general we will want to use more chern classes/coefficients of the Hilbert polynomial.

Remark 1.8. As a motto, we might say there are only injections to semistable objects in the counterclockwise direction. Dually, there are only surjections in the clockwise directions, since given a surjection $f : A \rightarrow B$ from a semistable A , we get

$$0 \rightarrow \ker f \rightarrow A \rightarrow B \rightarrow 0,$$

so $Z(A) = Z(B) + Z(\ker f)$, and by the semistability of A and $\ker f \leq A$ a subobject we know $\arg(\ker f) \leq \arg(A)$. Then, by forming a parallelogram at $0, Z(B), Z(\ker f), Z(A)$, we can see $\arg(B) \geq \arg(A)$.

Even better, between two semistable objects there are only *any* maps in the counterclockwise direction, as in the following proposition.

Proposition 1.9. If A, A' are semistable with $\theta > \theta'$, then $\text{Hom}(A, A') = 0$.

Proof. Let $f : A \rightarrow A'$, so we get SES

$$0 \rightarrow \ker f \rightarrow A \rightarrow \text{im} f \rightarrow 0.$$

Suppose to the contrary $\text{im} f \neq 0$. By the semistability of A we get $\arg(\text{im} f) \geq \arg(A)$, and by the the semistability of A' we get $\arg(\text{im} f) \leq \arg(A')$. So,

$$\arg(A) \leq \arg(\text{im} f) \leq \arg(A'),$$

but this contradicts $\theta > \theta'$. □

Proposition 1.10. If A, A' are stable with $\theta = \theta'$, then any nonzero morphism is an isomorphism. Furthermore, if $\mathcal{A}$ is k -linear (i.e., enriched over finite dimensional k vector spaces) for $k = \bar{k}$, then $\text{Hom}(A, A) = \text{id}_A$.

Proof. Let $f : A \rightarrow A'$ be a nonzero morphism. Then $\ker f \leq A$ must be trivial, otherwise it destabilizes A . Likewise, $\text{im} f \leq A'$ must be all of A' for the same reason.

Now, assume $\mathcal{A}$ is k -linear, and let $f : A \rightarrow A$. Then, $f_* : \text{End}(A) \rightarrow \text{End}(A)$ is a linear operator on a finite dimensional vector space. By Cayley-Hamilton, it is killed by some minimal polynomial $p(T) \in k[T]$, which has a root λ because k is algebraically closed, so

$$p(T) = (T - \lambda)q(T).$$

Since $q(T), T - \lambda$ are smaller degree polynomials, they do not kill f_* . If we evaluate our polynomials at f_* , we get

$$0 = (f_* - \lambda \text{id}_{\text{End}(A)})q(f_*),$$

and since $q(f_*) \neq 0$, we have $q(f) \neq 0$, so by the first statement of the proposition, $q(f)$ is an isomorphism, so $q(f_*)$ is invertible. So, we get

$$0 = f_* - \lambda \text{id}_{\text{End}(A)},$$

so $f = \lambda \text{id}_A$. □

Remark 1.11. The two propositions we proved is essentially a “half” Schur’s lemma from representation theory. Between stable objects A, A' , either

- they are isomorphic, and $\text{Hom}(A, A') \cong k$,
- $\theta(A) \geq \theta(A')$ and they are not isomorphic, and $\text{Hom}(A, A') = 0$.

Definition 1.12. A stability function $Z : K(\mathcal{A}) \rightarrow \mathbb{C}$ is a **pre-stability condition** if

- (i) $\mathcal{A}_\theta$ is Artinian (DCC)
- (ii) every object $A \in \mathcal{A}$ has a (unique) **Harder-Narasimhan filtration**

$$0 = A_0 \subseteq A_1 \subseteq \cdots \subseteq A_r \subseteq A_{r+1} = A$$

such that

- Associated graded components $G_i = A_{i+1}/A_i$ are semistable.
- Phases decrease, i.e.,

$$\arg(G_i) > \arg(G_{i+1}).$$

Remark 1.13. The Harder-Narasimhan filtration should be thought of as a Jordan-Holder sort of filtration—we should be able to refine our filtration until we get semistable associated graded pieces.

But, here we get something even better by requiring semistability with decreasing phases, because it gives us something canonical, because semistability controls morphisms into graded pieces.

Example 1.14 (Non-Canonical Filtration). As an example of what can “go wrong” in a filtration, for an abelian group A , we have a direct sum decomposition $A = A_{\text{tor}} \oplus A_{\text{free}}$, where $A_{\text{tor}} \leq A$ is canonically defined, but $A_{\text{free}} \leq A$ is not. For

$$A = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z},$$

we could take A_{free} to be either the span $\langle (1, 0) \rangle$ or $\langle (1, 1) \rangle$, so there’s no suitable sense in which the filtration

$$0 \leq A_{\text{free}} \leq A$$

is canonical. The “reason” for the ambiguity is that there is a nontrivial morphism $A_{\text{free}}/0 \rightarrow A/A_{\text{free}}$ between associated graded pieces

$$\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}, \quad 1 \mapsto 1$$

which allows us to “move” A_{free} .

Proposition 1.15. Harder-Narasimhan filtrations are unique.

Proof. Let $C \in \mathcal{A}$, and

$$\begin{aligned} 0 = A_0 &\leq A_1 \leq \cdots \leq A_r \leq A_{r+1} = C \\ 0 = B_0 &\leq B_1 \leq \cdots \leq B_s \leq B_{s+1} = C \end{aligned}$$

be two Harder-Narasimhan filtrations. Without loss of generality, assume $\arg(A_1) \geq \arg(B_1)$. Our strategy is to show that the map $A_1 \hookrightarrow C = B_{s+1}$ factors through as an injection to B_s , then to B_{s-1} , and so on until it factors through to B_1 .

We have the composition

$$A_1 \hookrightarrow B_{s+1} \rightarrow B_{s+1}/B_s$$

a morphism of semistable objects, where A_1 has greater phase, since $\arg(A_1/0) \geq \arg(B_1/0) \geq \cdots \geq \arg(B_{s+1}/B_s)$. So, by the proposition, the morphism is 0, so $A_1 \leq B_s$. Continuing inductively, we get $A_1 \leq B_1$.

Then, we must have $\arg(A_1) = \arg(B_1)$, since otherwise A_1 destabilizes B_1 . Now that the phases are equal, we can repeat the argument reversing the roles of A_i, B_i to get $B_1 \leq A_1$, so indeed $A_1 = B_1$.

Now, we can mod the filtrations out by $A_1 = B_1$, and repeat our argument on C/A_1 to get $A_2 = B_2$, and continue inductively. $\square$

Example 1.16. Our pre-stability condition on $\text{Coh}(\mathbb{P}^1)$ has $\mathcal{A}_\theta$ Artinian, since either θ is a rational angle and any descending filtration will terminate at a line bundle $\mathcal{O}(a)$ or at $\mathbb{C}_p$, and otherwise θ is irrational and $\mathcal{A}_\theta = 0$.

We also have Harder-Narasimhan filtrations in our example—given $\mathcal{F}$, set $A_1 = \mathcal{F}_{\text{tor}}$, and then slowly add all direct summand line bundles of highest degree.

So, we indeed have a pre-stability condition.

1.2 Failure for Surfaces

Now let's try to find a pre-stability condition for a surface S , where we mean a group homomorphism

$$Z : K(\text{Coh}(S)) \xrightarrow{\text{ch}} \mathbb{Z}^3 \rightarrow \mathbb{C}$$

sending $\mathcal{F} \neq 0$ to a point with $\theta \in (0, 1]$.

We're going to show that none exist. In later sections, we'll see that we have to pass to the derived category of $\text{Coh}(S)$ and adjust our definitions in order to actually have pre-stability conditions.

To show none exist, let C be a curve on S , and $\mathcal{O}_C$ the (pushforward of its) structure sheaf. Let P be a point on C , and $\mathcal{O}_P$ the corresponding skyscraper sheaf.

Suppose, to the contrary, there is a pre-stability condition given by Z , so in particular it will send $\mathcal{O}_P$, $\mathcal{O}_C$, and $\mathcal{O}_S$ to points in $\mathbb{C}$ with $\arg \in (0, 1]$.

- $\text{Im}(Z([\mathcal{O}_P]) = 0$, because otherwise eventually

$$\cdots \leq \mathcal{I}_{2P} \leq \mathcal{I}_P \leq \mathcal{O}_S \quad (*)$$

will not lie in the upper half plane, since $[\mathcal{I}_{nP}] = [\mathcal{O}_S] - n[\mathcal{O}_P]$ by writing down a short exact sequence. We also therefore have $\text{Re}(Z([\mathcal{O}_P])) < 0$.

- $\text{Im}(Z([\mathcal{O}_S])) > 0$, because otherwise either the chain $(*)$ would eventually make $\text{Re}(Z([\mathcal{I}_{nP}])) > 0$.
- Finally, we have to place $\mathcal{O}_C$.

- If $\text{Im}(\mathcal{O}_C) = 0$, then

$$\cdots \leq \mathcal{O}_C(-2P) \leq \mathcal{O}_C(-P) \leq \mathcal{O}_C$$

would eventually have $\text{Re } Z > 0$.

- If $\text{Im}(\mathcal{O}_C) > 0$, then

$$\cdots \leq \mathcal{O}_S(-2C) \leq \mathcal{O}_S(-C) \leq \mathcal{O}_S$$

would eventually have $\text{Im } Z < 0$.

So, we get a contradiction, and Z doesn't exist!

2 Homological Algebra

To remedy the failure of the existence of stability conditions on surfaces, we're going to have to “use an alternative t-structure on the derived category”. We'll do some work to get to these definitions, but essentially the story is the following.

- a stability condition decomposes an abelian category into semistables
- an abelian category $\mathcal{A}$ in turn decomposes its derived category $D(\mathcal{A})$.

So, in total a stability condition gives a very fine decomposition of the derived category. For surfaces, we will still want to decompose the derived category into semistables, but we will fail to do it in these steps. Instead, we will

- decompose the derived category $D(\mathcal{A})$ with an alternate abelian category, which is “the heart of a t-structure”.
- decompose the heart by giving it a stability condition.

2.1 Derived and Triangulated Categories

Now, we give a rapid tour of the required homological algebra background.

Definition 2.1. Let $\mathcal{A}$ be an abelian category.

- Denote $\text{Kom}(\mathcal{A})$ the **category of chain complexes** valued in $\mathcal{A}$, with chain maps as morphisms.
- Denote $K(\mathcal{A})$ the **homotopy category** of $\mathcal{A}$, which has the same objects as $\text{Kom}(\mathcal{A})$, but morphisms are quotiented by chain homotopy.
- Denote $D(\mathcal{A})$ the **derived category** of $\mathcal{A}$, which is $K(\mathcal{A})[\text{q-isos}^{-1}]$, i.e., the category with the same objects as $K(\mathcal{A})$ but with formal inverses to quasi-isomorphisms adjoined.

Remark 2.2. A few remarks are in place:

- We will always use cohomology indexing for chain complexes.
- We can put superscripts $+$, $-$, b on Kom , K or D to denote the category where we only have objects which are chain complexes with cohomology vanishing sufficiently far on the left, right, or both.

2.1.1 Morphisms in the Derived Category

In general, it is difficult to describe the morphisms of a category where we formally inverted morphisms (called **localization**, in analogy to the ring setting). However, in $K(\mathcal{A})$, the class of quasi-isomorphisms form a **localizing class** (I won't include the definition here) which makes the following true.

Definition 2.3. A **roof** in $D(\mathcal{A})$ is a diagram of the form

$$\begin{array}{ccc} & Z^\bullet & \\ s \swarrow & & \searrow f \\ X^\bullet & & Y^\bullet \end{array}$$

for s a quasi-isomorphism, and f a chain map (or rather both should be homotopy classes of maps), which should be thought of as $f \circ s^{-1} = \frac{f}{s}$. The roofs $X^\bullet \xleftarrow{s} Z^\bullet \xrightarrow{f} Y^\bullet$ and $X^\bullet \xleftarrow{t} W^\bullet \xrightarrow{g} Y^\bullet$ are **equivalent** if there exists a “common denominator” roof, i.e., there is a roof $Z^\bullet \xleftarrow{u} V^\bullet \xrightarrow{h} W^\bullet$ make the diagram commute (in $K(\mathcal{A})$, so up to homotopy)

$$\begin{array}{ccccc} & & V^\bullet & & \\ & u \swarrow & & \searrow h & \\ & Z^\bullet & & W^\bullet & \\ s \swarrow & & & & \searrow g \\ X^\bullet & & & & Y^\bullet \\ & \nwarrow t & & \nearrow f & \end{array}$$

which will at least imply $f \circ s^{-1} = g \circ t^{-1}$, since

$$f \circ s^{-1} = f \circ u \circ u^{-1} \circ s^{-1} = g \circ h \circ u^{-1} \circ s^{-1}$$

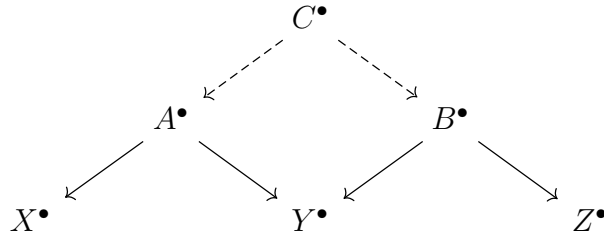
(the long route), and to get $g \circ t^{-1}$ to be the long route, we first notice

$$\begin{aligned} s \circ u &= t \circ h \\ t^{-1} &= h \circ s^{-1} \circ u^{-1} \end{aligned}$$

so with this identity

$$g \circ t^{-1} = g \circ h \circ s^{-1} \circ u^{-1}.$$

We can compose roofs $X^\bullet \leftarrow A^\bullet \rightarrow Y^\bullet$ and $Y^\bullet \leftarrow B^\bullet \rightarrow Z^\bullet$ by putting a roof over both of them.



This is always be possible, exercise for the reader. You should do this by studying cones/distinguished triangles.

Theorem 2.4.

$$\mathrm{Hom}_{D(\mathcal{A})}(X^\bullet, Y^\bullet) = \{\text{roofs } X^\bullet \text{ to } Y^\bullet\} / \text{equivalence},$$

i.e., all morphisms in the derived category are roofs.

In some cases, we can describe the morphisms even more explicitly. There is a functor $\mathcal{A} \rightarrow D(\mathcal{A})$ by sending an object to a degree 0 complex $A \mapsto (\cdots \rightarrow 0 \rightarrow A \rightarrow 0 \rightarrow \cdots)$. We will often identify $A \in \mathcal{A}$ with its degree 0 complex. We can then ask the following:

- Is $\mathcal{A} \rightarrow D(\mathcal{A})$ full? faithful? That is for $A, B \in \mathcal{A}$, can we describe

$$\mathrm{Hom}_{D(\mathcal{A})}(A, B)$$

in relation to $\mathrm{Hom}_{\mathcal{A}}(A, B)$?

- Can we understand $\mathrm{Hom}_{D(\mathcal{A})}(A, B[i])$?

For the second question, at least when $i > 0$, we have the following strategy for writing down morphisms. Take an exact sequence

$$0 \rightarrow B \rightarrow K^{-i+1} \rightarrow \cdots \rightarrow K^0 \rightarrow A \rightarrow 0,$$

which we denote by $K^\bullet$. In this case, we have the morphisms of chain complexes

$$\begin{array}{ccccccccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & A & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & & & & & & & & & \uparrow & & & & \\ 0 & \longrightarrow & B & \longrightarrow & K^{-i+1} & \longrightarrow & \cdots & \longrightarrow & K^{-1} & \longrightarrow & K^0 & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & & & & & & & & & & & \\ 0 & \longrightarrow & B & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

where the top morphism is a quasi-isomorphism (so has a formal inverse). Denote the middle complex by $\tilde{K}$, so that in total we get a morphism $A \rightarrow B[i]$ which is a roof.

$$\begin{array}{ccc} & \tilde{K} & \\ \swarrow & & \searrow \\ A & & B[i] \end{array}$$

Denote this morphism by $y(K^\bullet)$. We can then ask whether all morphisms $A \rightarrow B[i]$ arise in this way, and when are two such morphisms equivalent.

These questions are then answered by the following proposition.

- Proposition 2.5.** (a) $\text{Hom}_{D^*(\mathcal{A})}(X, Y[i]) = 0$ for $i < 0$
 (b) $\text{Hom}_{D^*(\mathcal{A})}(X, Y) = \text{Hom}_{\mathcal{A}}(X, Y)$
 (c) Any element of $\text{Hom}_{D^*(\mathcal{A})}(X, Y[i])$ is of the form $y(K^\bullet)$, and the composition corresponds to concatenating extensions.

Proof of (a). Let $\varphi : X \rightarrow Y[-i]$ for $i > 0$ be written as a roof $X \xleftarrow{s} K^\bullet \xrightarrow{f} Y[-i]$. To show $\varphi = 0$, we □

We also have the following theorem to help us understand morphisms: **TODO**

Proposition 2.6. $K^+(I) \rightarrow D^+(\mathcal{A})$ and $K^-(P) \rightarrow D^-(\mathcal{A})$ form equivalence of categories (assuming enough projectives/injectives). Consequently,

$$\text{Hom}_{K(\mathcal{A})}(X^\bullet, Y^\bullet) \cong \text{Hom}_{D(\mathcal{A})}(X^\bullet, Y^\bullet)$$

when either $Y^\bullet \in I$ or $X^\bullet \in P$.

Proof. **both of these would be good to understand!!!** □

2.1.2 Triangles

The derived category is usually no longer abelian, but it does have some extra structure. It's still additive (add roofs by expressing both as $f \circ s^{-1}, g \circ s^{-1}$ for some s , and then $(f + g) \circ s^{-1}$ is the sum), it still has the (additive) shift functor $X^\bullet \mapsto X^\bullet[1]$, and it has a notion of “distinguished triangles”, which somewhat analogous to exact sequences.

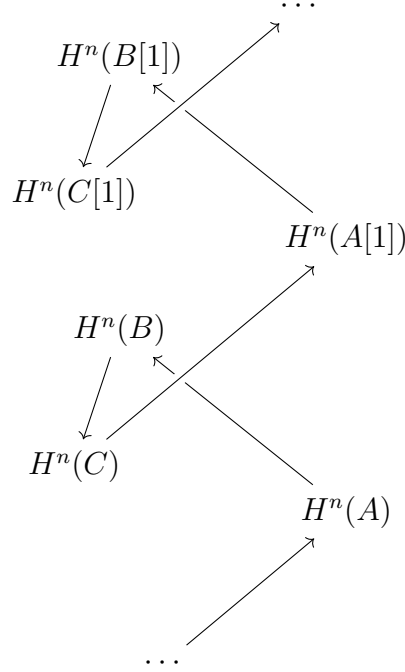
Recall that a short exact sequence

$$0 \rightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \rightarrow 0 \tag{*}$$

of chain complexes in $\text{Kom}(\mathcal{A})$ induces a long exact sequence in cohomology.

$$H^n(A^\bullet) \xrightarrow{H^n(f)} H^n(B^\bullet) \xrightarrow{H^n(g)} H^n(C^\bullet) \xrightarrow{\delta^n} H^{n+1}(A^\bullet) = H^n(A^\bullet[1]),$$

where the maps $H^n(f), H^n(g)$ are directly induced by f, g , but δ^n is maybe more mysterious. In any case, the LES might be drawn as a spiraling triangular helix.



One thing the derived category does for you is it makes it so δ^n is actually induced by a morphism $C \rightarrow A[1]$, so that this spiral is true not just on the level of cohomology, but on the level of the derived category.

More precisely, it is true that the short exact sequence $(*)$ is quasi-isomorphic to a short exact sequence

$$0 \rightarrow A^\bullet \rightarrow \text{Cyl}(f) \rightarrow C(f) \rightarrow 0,$$

where $\text{Cyl}(f), C(f)$ are the **mapping cylinder/mapping cone** respectively, analogous to topology, but they have a chain complex definition. In some sense, they are defined exactly the same as in topology—define the path object chain complex $I^\bullet$ by $I^{-1} = \mathbb{Z}e$, $I^0 = \mathbb{Z}a \oplus \mathbb{Z}b$, and $d^{-1}(e) = b - a$. Then, $\text{Cyl}(f)$ is the pushout of a diagram $A^\bullet \otimes I^\bullet \leftarrow A^\bullet \rightarrow B^\bullet$, and $C(f)$ is a quotient of $\text{Cyl}(f)$.

Also, like topology, there is a map

$$C(f) \xrightarrow{\delta} SA^\bullet := A^\bullet[1]$$

where we $SA^\bullet$ is the suspension (this map is “contracting $B^\bullet$ ”). This map induces the connecting map in the long exact sequence in cohomology, i.e.,

$$H^n(C(f)) \xrightarrow{H^n(\delta)} H^n(A^\bullet[1]) = H^{n+1}(A^\bullet)$$

is the connecting map. We will call a sequence of the form

$$A^\bullet \rightarrow \text{Cyl}(f) \rightarrow C(f) \rightarrow A^\bullet[1]$$

(or anything quasi-isomorphic to it) a **distinguished triangle** in $D(\mathcal{A})$. What we just discussed shows that any short exact sequence in $\text{Kom}(\mathcal{A})$ gives a distinguished triangle in $D(\mathcal{A})$.

Just as a short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ tells us that B is C extended by A , a distinguished triangle also tells us this sort of gluing information but with added symmetry. We axiomatize the formal structure of the derived category with the following notion.

Definition 2.7. A **triangulated category** Δ is an additive category with

- An additive automorphism $(-)[1] : \Delta \rightarrow \Delta$ called the **shift/translation/suspension functor** (some authors relax to auto-equivalence, but this is tricky).
- A collection of triangles

$$\{A^\bullet \rightarrow B^\bullet \rightarrow C^\bullet \rightarrow A^\bullet[1]\}$$

called distinguished triangles

such that the following axioms hold:

- **TR1:**

- Every triangle isomorphic to a distinguished triangle is itself a distinguished triangle.
- The triangle

$$X \xrightarrow{\text{id}_X} X \rightarrow 0 \rightarrow X[1]$$

is a distinguished triangle.

- Every morphism $f : X \rightarrow Y$ can be completed to a distinguished triangle

$$X \xrightarrow{f} Y \rightarrow C(f) \rightarrow X[1],$$

where $C(f)$ is called the **cone** of f . The other axioms will imply $C(f)$ is unique up to (non-unique) isomorphism.

- **TR2:** a triangle

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

is distinguished iff

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-T(f)} Y[1]$$

is distinguished (we can “rotate” distinguished triangles, giving us the spiraling triangular helix we drew earlier).

- **TR3:** Given a diagram of solid arrows

$$\begin{array}{ccccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & X[1] \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow T(\alpha) \\ X' & \xrightarrow{f'} & Y' & \xrightarrow{g'} & Z' & \xrightarrow{h'} & X'[1] \end{array}$$

where the rows are distinguished triangles and the left square commutes, then there exists $\gamma : Z \rightarrow Z'$ making the entire diagram commute.

- **TR4 (octahedral axiom):** Given three distinguished triangles of the form

$$\begin{array}{ccccccc}
X & \xrightarrow{f} & Y & \xrightarrow{p} & Y/X & \longrightarrow & X[1] \\
Y & \xrightarrow{g} & Z & \xrightarrow{q} & Z/Y & \longrightarrow & Y[1] \\
X & \xrightarrow{g \circ f} & Z & \xrightarrow{r} & Z/X & \longrightarrow & X[1]
\end{array}$$

(where we are denoting the cones suggestively like quotients), there exists a distinguished triangle

$$Y/X \xrightarrow{\bar{g}} Z/X \xrightarrow{\pi} Z/Y \xrightarrow{s} (Y/X)[1]$$

such that the following diagram commutes

$$\begin{array}{ccccccc}
X & \xrightarrow{g \circ f} & Z & \xrightarrow{q} & Z/Y & \xrightarrow{s} & (Y/X)[1] \\
& \searrow f & \nearrow g & \searrow r & \nearrow \pi & \searrow & \nearrow p[1] \\
& & Y & & Z/X & & Y[1] \\
& & \searrow p & \nearrow \bar{g} & \searrow & \nearrow f[1] & \\
& & & Y/X & \longrightarrow & & X[1]
\end{array}$$

Remark 2.8. The octahedral axiom is roughly a version of the “third isomorphism theorem”, which says $(Z/X)/(Y/X) \cong Z/Y$. The commuting diagram says something about how the projections/suspensions interact.

Remark 2.9. In TR3, we do *not* assert that γ is unique, just that some γ exists. In fancier words, the cone construction is not functorial.

Theorem 2.10. For $\mathcal{A}$ and abelian category, $D^*(\mathcal{A})$ (where $*$ $\in \{+, -, b, \}$) is a triangulated category with the shift functor and the described distinguished triangles.

We will not prove this in these notes, but I don’t believe it is particularly hard.

Now, we’re going to give ourselves a little bit of scaffolding for how to handle a triangulated category. We’re not going to be very comprehensive—go to the Stacks’ project if that’s what you want.

Proposition 2.11. In a distinguished triangle

$$X \xrightarrow{f} Y \xrightarrow{g} Z \rightarrow X[1],$$

$$g \circ f = 0.$$

Proof. By TR1 we have the distinguished triangle $X \xrightarrow{\text{id}_X} X \rightarrow 0 \rightarrow X[1]$. We can then apply TR3 to get the following commutative diagram

$$\begin{array}{ccccccc}
X & \xrightarrow{\text{id}_X} & X & \longrightarrow & 0 & \longrightarrow & X[1] \\
\downarrow \text{id}_X & & \downarrow f & & \downarrow & & \downarrow \text{id}_{X[1]} \\
X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & X[1]
\end{array}$$

so indeed $g \circ f = 0$. □

This tells us, at least a little bit, distinguished triangles behave like exact sequences.

Definition 2.12. Let Δ a triangulated category, and $\mathcal{A}$ an abelian category. A **cohomological functor** is a functor $H : \Delta \rightarrow \mathcal{A}$ such that every distinguished triangle

$$X \rightarrow Y \rightarrow Z \rightarrow X[1]$$

is sent to an exact sequence

$$H(X) \rightarrow H(Y) \rightarrow H(Z)$$

(and by TR2, we can continue this exact sequence in a spiral).

Example 2.13. When $\Delta = D^*(\mathcal{A})$, the usual cohomology functor H^n on $\text{Kom}(\mathcal{A})$ descends to a functor on $D^*(\mathcal{A})$ (almost by definition), and indeed, the cohomology functor is a cohomological functor.

Example 2.14. Let Δ be a triangulated category. In particular, Δ is additive, so the hom-sets are abelian groups. So, for any $W \in \Delta$, we can consider the hom functor as

$$\text{Hom}(W, -) : \Delta \longrightarrow \mathbb{Z}\text{-Mod}.$$

We claim this is a cohomological functor. Let

$$X \xrightarrow{f} Y \xrightarrow{g} Z \rightarrow X[1]$$

be distinguished, we show exactness at the middle term

$$\text{Hom}(W, X) \xrightarrow{\text{Hom}(W, f)} \text{Hom}(W, Y) \xrightarrow{\text{Hom}(W, g)} \text{Hom}(W, Z).$$

We at least have $\text{Hom}(W, g \circ f) = 0$, since $g \circ f = 0$. Now, let $y : W \rightarrow Y$ such that $g \circ y = 0$. We wish to show $y = f \circ x$. To see this, rotate with TR2, apply TR3, and rotate back to get the commutative diagram

$$\begin{array}{ccccccc} W & \xrightarrow{\text{id}_W} & W & \longrightarrow & 0 & \longrightarrow & W[1] \\ \downarrow x & & \downarrow y & & \downarrow 0 & & \downarrow x[1] \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & X[1] \end{array}$$

so indeed, we get exactness.

Proposition 2.15. Cones are unique up to isomorphism.

Proof. This will essentially be the 5-lemma. Suppose

$$X \rightarrow Y \rightarrow Z \rightarrow X[1], \quad X \rightarrow Y \rightarrow Z' \rightarrow X[1]$$

are both distinguished triangles. Then, applying TR3, we get $\gamma : Z \rightarrow Z'$ giving us a morphism between the distinguished triangles. To show γ is an isomorphism it suffices, by Yoneda, to show for all $W \in \Delta$, $\text{Hom}(W, \gamma)$ is an isomorphism. Indeed, applying the functor $\text{Hom}(W, -)$ to the diagram which is the morphism of distinguished triangles, we get

$$\begin{array}{ccccccccc}
\mathrm{Hom}(W, X) & \rightarrow & \mathrm{Hom}(W, Y) & \rightarrow & \mathrm{Hom}(W, Z) & \rightarrow & \mathrm{Hom}(W, X[1]) & \rightarrow & \mathrm{Hom}(W, Y[1]) \\
\downarrow (id_X)_* & & \downarrow (id_Y)_* & & \downarrow \gamma_* & & \downarrow (id_{X[1]})_* & & \downarrow (id_{Y[1]})_* \\
\mathrm{Hom}(W, X) & \rightarrow & \mathrm{Hom}(W, Y) & \rightarrow & \mathrm{Hom}(W, Z') & \rightarrow & \mathrm{Hom}(W, X[1]) & \rightarrow & \mathrm{Hom}(W, Y[1])
\end{array}$$

where all vertical morphisms (except for potentially the middle one) are isomorphisms, so by the 5-lemma the middle is an isomorphism. $\square$

Proposition 2.16. The direct sum of distinguished triangles in distinguished, i.e., if

$$X_i \rightarrow Y_i \rightarrow Z_i \rightarrow X_i[1] \quad (S_i)$$

is distinguished for $i = \{1, 2\}$, then

$$X_1 \oplus X_2 \rightarrow Y_1 \oplus Y_2 \rightarrow Z_1 \oplus Z_2 \rightarrow X_1[1] \oplus X_2[1]$$

Proof. We at least have distinguished triangle

$$X_1 \oplus X_2 \rightarrow Y_1 \oplus Y_2 \rightarrow Z \rightarrow X_1[1] \oplus X_2[1], \quad (T)$$

and we want to show that this is isomorphic to our “desired” triangle. We have morphisms $S_i \rightarrow T$ by completing the commutative square

$$\begin{array}{ccc}
X_i & \longrightarrow & Y_i \\
\downarrow & & \downarrow \\
X_1 \oplus X_2 & \longrightarrow & Y_1 \oplus Y_2
\end{array}$$

to a distinguished triangle. Now, we want to argue that $S_1 \oplus S_2 \rightarrow T$ is an isomorphism. We do this by hitting the morphism of distinguished triangles with $\mathrm{Hom}(W, -)$. Even though $S_1 \oplus S_2$ is not a-priori distinguished, it will still turn into an exact sequence after being hit with $\mathrm{Hom}(W, -)$, since it will be a direct sum of exact sequences $\mathrm{Hom}(W, S_i)$. Therefore, we can finish the proof by applying the 5-lemma. $\square$

Proposition 2.17. The morphism $A \xrightarrow{0} B$ is completed to a distinguished triangle via

$$A \xrightarrow{0} B \xrightarrow{i} A[1] \oplus B \xrightarrow{\pi} A[1],$$

where i, π are inclusions/projections into the direct sum.

Proof. We have distinguished triangles

$$A \xrightarrow{id_A} A \rightarrow 0 \rightarrow A[1],$$

which rotates via TR3 to

$$A \rightarrow 0 \rightarrow A[1] \xrightarrow{id_{A[1]}} A[1].$$

Likewise, we have distinguished triangle

$$0 \rightarrow B \xrightarrow{\text{id}_B} B \rightarrow 0[1].$$

Their direct sum is a distinguished triangle

$$A \xrightarrow{0} B \xrightarrow{i} A[1] \oplus B \xrightarrow{\pi} A[1].$$

□

We will define/develop some helpful notation.

Definition 2.18. Let $A, C \in \mathcal{D}$. We call B an **extension** of A, C if there is a distinguished triangle

$$A \rightarrow B \rightarrow C \rightarrow A[1].$$

Let $E_1, \dots, E_n \in \mathcal{D}$ be some objects (resp. $\mathcal{C} \subseteq \mathcal{D}$ a full subcategory). Denote the **extension closure**

$$\langle E_1, \dots, E_n \rangle \subseteq \mathcal{D}, \quad (\text{resp. } \langle \mathcal{C} \rangle \subseteq \mathcal{D})$$

the smallest full subcategory containing $E_1, \dots, E_n$ (resp. $\mathcal{C}$) which is closed under extensions.

Definition 2.19. Let $\mathcal{C} \subseteq \mathcal{D}$ be a full subcategory of a triangulated category $\mathcal{D}$. Denote

$${}^\perp\mathcal{C} = \{X \in \mathcal{D} \mid \text{Hom}(X, \mathcal{C}) = 0\},$$

where by $\text{Hom}(X, \mathcal{C}) = 0$, we mean for all $C \in \mathcal{C}$, $\text{Hom}(X, C) = 0$. Similarly, define

$$\mathcal{C}^\perp = \{X \in \mathcal{D} \mid \text{Hom}(\mathcal{C}, X) = 0\}.$$

If we squint our eyes and think of Hom as some sort of non-symmetric inner product, then we are exactly looking at the (one-sided) orthogonal complements. We put the $\perp$ on the side of $\mathcal{C}$ where we should put X to get Hom with $\mathcal{C}$ to be 0.

Proposition 2.20. ${}^\perp\mathcal{C}$ and $\mathcal{C}^\perp$ are closed under extensions.

Proof. Let $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ be a distinguished triangle with $X, Z \in {}^\perp\mathcal{C}$. We want to show $Y \in {}^\perp\mathcal{C}$, so for an arbitrary object $C \in \mathcal{C}$, we show $\text{Hom}(Y, C) = 0$.

To do this, we hit the triangle with $\text{Hom}(-, C)$ to get the exact sequence

$$\text{Hom}(X, C) \leftarrow \text{Hom}(Y, C) \leftarrow \text{Hom}(Z, C),$$

and by assumption the left and right terms are 0, so $\text{Hom}(Y, C) = 0$. This shows ${}^\perp\mathcal{C}$ is closed under extensions.

An analogous proof shows $\mathcal{C}^\perp$ is also closed under extensions. □

Corollary 2.21. Let $\mathcal{C}, \mathcal{C}'$ be full subcategories of $\mathcal{D}$. If $\mathcal{C}' \subseteq {}^\perp\mathcal{C}$, then $\langle \mathcal{C}' \rangle \subseteq {}^\perp\mathcal{C}$.

Proposition 2.22 (Verdier’s Exercise/The 3×3 lemma). Assume

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z & \longrightarrow & W \end{array}$$

is commutative. Then, there exists an object U and morphisms making all rows and columns distinguished triangles.

$$\begin{array}{ccccccc} X & \longrightarrow & Y & \longrightarrow & Y/X & \longrightarrow & X[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ Z & \longrightarrow & W & \longrightarrow & W/Z & \longrightarrow & Z[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ Z/X & \dashrightarrow & W/Y & \dashrightarrow & U & \dashrightarrow & (Z/X)[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ X[1] & \longrightarrow & Y[1] & \longrightarrow & (Y/X)[1] & \longrightarrow & X[2] \end{array}$$

Proof Sketch. We get $Y/X \rightarrow W/Z$ by TR1 and TR3, likewise for $Z/X \rightarrow W/Y$ (i.e., we get the diagram without the bottom 2×2 square, with rows and columns distinguished). Following TR1, we get two “options” for U , which we will denote respectively

$$(W/Z)/(Y/X), \quad (W/Y)/(Z/X).$$

Heuristically, if cross multiply we get the same formal fractions which gives us some hope they are isomorphic and that the squares commute.

□

2.2 t-structures

The derived category $D(\mathcal{A})$ of an abelian category $\mathcal{A}$ comes with even more structure called a t-structure—essentially this is the inclusion of $\mathcal{A} \hookrightarrow D(\mathcal{A})$ as degree 0 complexes.

Remark 2.23. We can understand $\mathcal{A} \hookrightarrow D(\mathcal{A})$ as the codomain category of H^0 . In fact, we can understand H^0 as *truncation*. Define $D(\mathcal{A})^{\leq 0}$ as the full subcategory of complexes $A^\bullet$ with $H^n(A^\bullet) = 0$ for $n > 0$, and $D(\mathcal{A})^{\geq 0}$ as those with $H^n(A^\bullet) = 0$ for $n < 0$.

Now, define the left truncation functor

$$\begin{aligned} \tau_{\geq 0} : D(\mathcal{A}) &\longrightarrow D(\mathcal{A})^{\geq 0} \\ \left(\cdots \rightarrow A^{-1} \xrightarrow{d^0} A^0 \xrightarrow{d^1} A^1 \rightarrow \cdots \right) &\longmapsto (\cdots \rightarrow 0 \rightarrow 0 \rightarrow \operatorname{coker} d^0 \rightarrow A^1 \rightarrow \cdots) \\ &\simeq (\cdots \rightarrow 0 \rightarrow \operatorname{im} d^{-1} \hookrightarrow A^0 \rightarrow A^1 \rightarrow \cdots) \end{aligned}$$

and the right truncation functor

$$\begin{aligned} \tau_{\leq 0} : D(\mathcal{A}) &\longrightarrow D(\mathcal{A})^{\leq 0} \\ \left(\cdots \rightarrow A^{-1} \xrightarrow{d^0} A^0 \xrightarrow{d^1} A^1 \rightarrow \cdots \right) &\longmapsto \left(\cdots \rightarrow A^{-1} \rightarrow \ker d^0 \rightarrow 0 \rightarrow 0 \rightarrow \cdots \right) \\ &\simeq \left(\cdots \rightarrow A^{-1} \rightarrow A^0 \twoheadrightarrow \operatorname{im} d^1 \rightarrow 0 \rightarrow \cdots \right) \end{aligned}$$

where these functors really are defined on the level of derived categories, i.e., quasi-isomorphic complexes give quasi-isomorphic truncations. Then, one can check for $i : \mathcal{A} \rightarrow D(\mathcal{A})$ the inclusion of degree 0 complexes,

$$i \circ H^0 \cong \tau_{\leq 0} \circ \tau_{\geq 0} \cong \tau_{\geq 0} \circ \tau_{\leq 0},$$

where the compositions commuting is the fact that for a complex $A \xrightarrow{f} B \xrightarrow{g} C$

$$\operatorname{coker}(A \xrightarrow{f'} \ker g) \cong \ker g / \operatorname{im} f \cong \ker(\operatorname{coker} f \xrightarrow{\bar{g}} C).$$

So, we'll see a t-structure as a way to give generalized cohomology. Other cohomologies besides H^0 are gotten just by shifting.

Remark 2.24. We can then refine our above ideas that cohomology is a way of giving objects a filtration in the derived category. Given an object $A^\bullet \in D(\mathcal{A})$, we have the **cohomology filtration**

$$0 \rightarrow \cdots \rightarrow \tau_{\leq n-1} A^\bullet \rightarrow \tau_{\leq n} A^\bullet \rightarrow \cdots \rightarrow A$$

where we have the short exact sequence

$$0 \rightarrow \tau_{\leq n-1} A^\bullet \rightarrow \tau_{\leq n} A^\bullet \rightarrow \left(\cdots \rightarrow 0 \rightarrow A^{n-1} / \ker d^n \rightarrow \ker d^{n+1} \rightarrow 0 \rightarrow \cdots \right) \rightarrow 0,$$

but in the derived category

$$\left(\cdots \rightarrow 0 \rightarrow A^{n-1} / \ker d^n \rightarrow \ker d^{n+1} \rightarrow 0 \rightarrow \cdots \right) \simeq \left(\cdots \rightarrow 0 \rightarrow 0 \rightarrow H^n(A^\bullet) \rightarrow 0 \rightarrow \cdots \right),$$

since there is a map left to right inducing isomorphisms of cohomology. So, in the derived category we have the triangle

$$\tau_{\leq n-1} A^\bullet \rightarrow \tau_{\leq n} A^\bullet \rightarrow H^n(A^\bullet)[-n] \rightarrow \tau_{\leq n-1} A^\bullet[1],$$

where the $-n$ shift is to place the cohomology at degree n . If we think of cones as a derived category analogy of quotients/kernels, we should view this as saying that $\tau_{\leq n} A^\bullet$ is built by gluing $H^n(A^\bullet)[-n]$ to $\tau_{\leq n-1} A^\bullet$.

In this way, the abelian category $\mathcal{A} \hookrightarrow D(\mathcal{A})$ is important because shifts of it (the cohomologies of a complex) are used to build the entire derived category by gluing. We will give two related definitions, which are equivalent in certain setups explained below.

2.2.1 Basics

This first definition defines a t-structure by defining the image of the truncation functors—by specifying $\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0}$, we will later define the truncations as adjoints to the inclusions of these categories (and the adjoints will exist).

Definition 2.25. Let $\mathcal{D}$ be a triangulated category. A **t-structure** on $\mathcal{D}$ is a pair $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ of strictly full subcategories (strictly=closed under isomorphisms), which satisfying some axioms. Denote $\mathcal{D}^{\leq n} := \mathcal{D}^{\leq 0}[-n]$, $\mathcal{D}^{\geq n} := \mathcal{D}^{\geq 0}[-n]$, and require

- (i) $\text{Hom}_{\mathcal{D}}(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 1}) = 0$.
- (ii) If $X \in \mathcal{D}^{\leq 0}$, then $X[1] \in \mathcal{D}^{\leq 0}$. If $Y \in \mathcal{D}^{\geq 0}$, then $Y[-1] \in \mathcal{D}^{\geq 0}$.
- (iii) For $A \in \mathcal{D}$, there exists a distinguished triangle

$$X \rightarrow A \rightarrow Y \rightarrow X[1]$$

such that $X \in \mathcal{D}^{\leq 0}$ and $Y \in \mathcal{D}^{\geq 1}$. These will be unique up to isomorphism, and we denote $\tau^{\leq 0}A := X$ and $\tau^{\geq 1}A := Y$

In particular, (i) implies $\mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 1} = 0$, because otherwise a nonzero object E would have the nonzero morphism id_E .

The t-structure is said to be **non-degenerate** if $\bigcap_{n \in \mathbb{N}} \mathcal{D}^{\leq n} = \bigcap_{n \in \mathbb{N}} \mathcal{D}^{\geq n} = \{0\}$. It is said to be **bounded** if for all $E \in \mathcal{D}$, there exists $n \in \mathbb{N}$ such that $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$. In particular, bounded t-structures are non-degenerate because a nonzero $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$ will *not* be in $\mathcal{D}^{\geq n+1}$ nor $\mathcal{D}^{\leq n-1}$.

We denote $\tau^{\leq n}A = \tau^{\leq 0}(A[n])[-n]$ and $\tau^{\geq n}A = \tau^{\leq 1}(A[n-1])[-n+1]$, called the **truncation functors** (we will soon prove they are well-defined and functorial).

The **heart** of the t-structure is $\mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$, denoted $\mathcal{D}^{\heartsuit}$. We will later see that the heart of a t-structure is abelian.

For $A \in \mathcal{D}$, we denote $H_{\mathcal{P}}^n(A) := (\tau^{\leq n}\tau^{\geq n}A)[n] \in \mathcal{D}^{\heartsuit}$, called the **cohomology** of A with respect to the t-structure. This name makes sense, because we will later prove that

$$H_{\mathcal{P}}^0 : \mathcal{D} \rightarrow \mathcal{D}^{\heartsuit}$$

is a cohomological functor.

Proposition 2.26. Let $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ be a t-structure. Then, $\tau^{\leq 0}, \tau^{\geq 1}$ (as well as their translates) are well-defined functors, and they form adjoint pairs

$$(i^{\leq 0}, \tau^{\leq 0}), \quad (\tau^{\geq 0}, i^{\geq 0})$$

Proof. To show the truncations are well-defined functors, we show that given $f : A \rightarrow A'$ and distinguished triangles

$$X \rightarrow A \rightarrow Y \rightarrow X[1], \quad X' \rightarrow A' \rightarrow Y' \rightarrow X'[1]$$

with $X, X' \in \mathcal{D}^{\leq 0}$ and $Y, Y' \in \mathcal{D}^{\geq 1}$, then we get a unique morphism of the distinguished triangles.

In particular, supposing we have shown this, we can apply this to $f = \text{id}_A$ twice to get a diagram

$$\begin{array}{ccccccc}
X & \longrightarrow & A & \longrightarrow & Y & \longrightarrow & X[1] \\
\downarrow & \searrow & \downarrow & \searrow & \downarrow & \searrow & \downarrow \\
X' & \longrightarrow & A & \longrightarrow & Y' & \longrightarrow & X'[1] \\
\downarrow & \searrow & \downarrow & \searrow & \downarrow & \searrow & \downarrow \\
X & \longrightarrow & A & \longrightarrow & Y & \longrightarrow & X[1]
\end{array}$$

and by uniqueness of the morphisms “extending” id_A , we are forced to have the composition be identity. This argument eventually tells us that the triangle given by the t-structure is unique up to isomorphism.

So, we just have to show this fact. We start by completing the diagram.

$$\begin{array}{ccc}
X & \longrightarrow & A \\
\downarrow & & \downarrow f \\
X' & \longrightarrow & A'
\end{array}$$

Since $\text{Hom}(X, -)$ is a cohomological functor, we have an exact sequence

$$\text{Hom}(X, Y'[-1]) \rightarrow \text{Hom}(X, X') \rightarrow \text{Hom}(X, A') \rightarrow \text{Hom}(X, Y'),$$

and we want the middle map an isomorphism, so that the composition $X \rightarrow A \xrightarrow{f} A'$ is the pushforward of a unique $X \rightarrow X'$. Now, this follows from the axioms—we have $Y', Y'[-1] \in \mathcal{D}^{\geq 0}$ whereas $X \in \mathcal{D}^{\leq 0}$, so the left and right terms in the exact sequence vanish. A similar argument gives a unique map $Y \rightarrow Y'$ making a square commute, and these morphisms must assemble into a map of distinguished triangles by TR3.

Now, we can indeed define the truncation functor $\tau^{\leq 0}$ just by picking for each A some isomorphism representative of X . We use a similar argument for $\tau^{\geq 1}$.

Finally, we say some words about the first adjunction. Given $X \in \mathcal{D}^{\leq 0}$ and $A \in \mathcal{D}$. Then, a morphism $X \rightarrow A$ in $\mathcal{D}$ gives a unique morphism $\tau^{\leq 0} X \rightarrow A$ and vice versa by the argument described before. This gives the bijection between the hom sets—one is left to check it is natural, and to work out the other adjunction. $\square$

2.2.2 Bounded t-structures

Now, we give an alternative definition of a bounded t-structure, which emphasizes the cohomology filtration.

Definition 2.27. A **bounded t-structure** is the data of a strictly full additive subcategory $\mathcal{A}^\sharp \leq \mathcal{D}$ such that

- (i) For $k_1 > k_2$, we have $\text{Hom}_{\mathcal{D}}(\mathcal{A}^\sharp[k_1], \mathcal{A}^\sharp[k_2]) = 0$ (“no negative Ext”).
- (ii) For every object E in $\mathcal{D}$ there are integers $k_1 > k_2 > \dots > k_n$ and a sequence (the top row)

$$0 = E^0 \longrightarrow E^1 \longrightarrow E^2 \rightarrow \dots \rightarrow E^{n-1} \longrightarrow E^n = E$$

where the cones over successive morphisms A^i are in $\mathcal{A}^\sharp[k_i]$. We will again call this sequence **t-structure cohomology filtration**.

Example 2.28. Given a t-structure cohomology filtration on E as above, we get a t-structure cohomology filtration on E/E^1 (the cone over $E^1 \rightarrow E$) by “quotienting” all the way through. That is, if we have the distinguished triangle

$$E^{i-1} \rightarrow E^i \rightarrow A^i \rightarrow E^{i-1}[1],$$

we should temporarily denote $E^i/E^{i-1} := A^i$, and write out the distinguished triangles (neglecting to write the third term)

$$\begin{aligned} E^1 &\rightarrow E^{i-1} \rightarrow E^{i-1}/E^1 \\ E^1 &\rightarrow E^i \rightarrow E^i/E^1, \end{aligned}$$

where $E^1 \rightarrow E^1$ is exactly the composite of $E^1 \rightarrow E^{i-1}$ and $E^{i-1} \rightarrow E^i$ (from the original distinguished triangle in the filtration). So, we are in the position to apply the octahedral axiom to get a distinguished triangle

$$E^{i-1}/E^1 \rightarrow E^i/E^1 \rightarrow E^i/E^{i-1} = A^i,$$

so indeed we get a sequence of morphisms $E^{i-1}/E^1 \rightarrow E^i/E^1$ which filters E using the same cones A^i . We did, however, throw away $E^1 = A^1$. We will use this construction to do induction on cohomology filtrations.

Proposition 2.29. The t-structure cohomology filtration, if we require all $A_i \neq 0$, is unique up to isomorphism.

Proof. Suppose

$$0 = F^0 \longrightarrow F^1 \longrightarrow F^2 \rightarrow \dots \rightarrow F^{m-1} \longrightarrow F^m = E$$

is another cohomology filtration, with $B_i \in \mathcal{A}^\sharp[\ell_i]$. Our strategy is to “pullback” the map $E \xrightarrow{\text{id}_E} E$ to the earliest respective objects in the filtration, then quotient out, and repeat.

Assume without loss of generality that $k_1 \geq \ell_1$. We can factor $E^1 \rightarrow E$ through F^1 , since we have distinguished triangle

$$F^1 \rightarrow E \rightarrow E/F^1 \rightarrow F^1[1], \quad (*)$$

where by E/F^1 has a t-structure cohomology filtration by only $B^2, \dots, B^m$, so

$$E/F^1 \in \langle \mathcal{A}^\sharp[\ell_2], \dots, \mathcal{A}^\sharp[\ell_m] \rangle \subseteq \mathcal{A}^\sharp[k_1]^\perp,$$

so in particular the composition $E^1 \rightarrow E \rightarrow E/F^1$ is zero. By applying $\text{Hom}(E^1, \cdot)$ to $(*)$, we get exact sequence

$$\text{Hom}(E^1, F^1) \rightarrow \text{Hom}(E^1, E) \rightarrow \text{Hom}(E^1, E/F^1),$$

so $E^1 \rightarrow E \rightarrow E/F^1$ being 0 implies we can pull back $E^1 \rightarrow E$ to $E^1 \rightarrow F^1$.

Now, if $k_1 > \ell_1$ strictly, then E_1 even factors through $E^0 = 0$, so $E^1 \rightarrow E$ is 0. Even better, we will get $E^1 \xrightarrow{\text{id}_{E^1}} E^1$ is zero, which would be a contradiction because $E^1 = A^1 \neq 0$. This is because we have the exact sequence

$$\text{Hom}(E^1, (E/E^1)[-1]) \rightarrow \text{Hom}(E^1, E^1) \rightarrow \text{Hom}(E^1, E),$$

with lefthand side 0, since

$$E/E^1 \in \langle \mathcal{A}^\sharp[k_2], \dots, \mathcal{A}^\sharp[k_n] \rangle \subseteq \mathcal{A}^\sharp[k_1]^\perp.$$

Therefore, we indeed have $k_1 = \ell_1$.

Now, we argue that $E^1 \rightarrow F^1$ is an isomorphism, by considering its cone, which we denote F^1/E^1 . We have the following distinguished triangles (neglecting to write the last term).

$$\begin{aligned} E^1 &\rightarrow F^1 \rightarrow F^1/E^1 \\ E^1 &\rightarrow E \rightarrow E/E^1 \\ F^1 &\rightarrow E \rightarrow E/F^1, \end{aligned}$$

where $E^1 \rightarrow E$ is the composite of $E^1 \rightarrow F^1 \rightarrow E$, so we are exactly in the position to apply the octahedral axiom to get the distinguished triangle

$$F^1/E^1 \rightarrow E/E^1 \rightarrow E/F^1,$$

but $F^1/E^1 \in \langle \mathcal{A}^\sharp[k_1], \mathcal{A}^\sharp[k_1 + 1] \rangle$ while $E/E^1 \in \langle \mathcal{A}^\sharp[k_2], \dots, \mathcal{A}^\sharp[k_n] \rangle$, and as usual there are no “negative Exts”, so $F^1/E^1 \rightarrow E/E^1$ is the zero morphism, so by Proposition 2.17,

$$E/F^1 \cong E/E^1 \oplus (F^1/E^1)[1],$$

but, if we assume $E^1 \rightarrow F^1$ is not an isomorphism, then $F^1/E^1 \neq 0$, so we get a nonzero morphism via the inclusion

$$(F^1/E^1)[1] \rightarrow E/E^1 \oplus (F^1/E^1)[1] \cong E/F^1,$$

which violates

$$E/F^1 \in \langle \mathcal{A}^\#[\ell_2], \dots, \mathcal{A}^\#[\ell_m] \rangle \subseteq \langle \mathcal{A}^\#[k_1], \mathcal{A}^\#[k_1 + 1] \rangle^\perp.$$

So, $E^1 \rightarrow F^1$ is an isomorphism, so we get $E/E^1 \rightarrow E/F^1$ an isomorphism via the 5-lemma from the following diagram.

$$\begin{array}{ccccccccc} E^1 & \longrightarrow & E & \longrightarrow & E/E^1 & \longrightarrow & E^1[1] & \longrightarrow & E[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ F^1 & \longrightarrow & E & \longrightarrow & E/F^1 & \longrightarrow & F^1[1] & \longrightarrow & E[1] \end{array}$$

Now, the quotiented filtrations E^i/E^1 , F^i/F^1 of $E/E^1 \cong E/F^1$ have strictly smaller n, m , so by induction they are isomorphic. To finish the argument, one should argue that the isomorphism of the quotiented filtrations lifts to an isomorphism of the original filtrations. $\square$

This next proposition is Lemma 3.2 in Bridgeland's original paper. He says it's "a good exercise in manipulating the definitions".

Proposition 2.30. Let $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ give a t-structure, such that for all $E \in \mathcal{D}$, there exists $n \in \mathbb{N}$ with $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$. Then, $\mathcal{A}^\# := \mathcal{D}^\heartsuit$ is a bounded t-structure in the other sense.

Conversely, given a bounded t-structure specified by $\mathcal{A}^\#$, define

$$\mathcal{D}^{\leq 0} = \{E \in \mathcal{D} \mid \text{all } A_i \neq 0 \text{ has } k_i \geq 0 \text{ in the t-structure cohomology filtration}\},$$

$$\mathcal{D}^{\geq 0} = \{E \in \mathcal{D} \mid \text{all } A_i \neq 0 \text{ has } k_i \leq 0 \text{ in the t-structure cohomology filtration}\},$$

then, $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ forms a t-structure such that $\mathcal{D}^\heartsuit = \mathcal{A}^\#$ and for all $E \in \mathcal{D}$, there exists $n \in \mathbb{N}$ with $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$.

Proof. We show that for $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ as in the first definition, $\mathcal{A}^\# := \mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$ satisfies the second definition. It's clear that

$$\text{Hom}_{\mathcal{D}}(\mathcal{A}^\#[k_1], \mathcal{A}^\#[k_2]) = \text{Hom}_{\mathcal{D}}(\mathcal{A}^\#, \mathcal{A}^\#[k_2 - k_1]) = 0$$

since $\mathcal{A}^\# \subseteq \mathcal{D}^{\leq 0}$, $\mathcal{A}^\#[k_2 - k_1] \subseteq \mathcal{D}^{\geq 1}$. Now, we get the filtration as follows. Take $\tau^{\leq n} E$, and by (iii) in the first definition, we get a distinguished triangle (now naming the objects with the truncation functors)

$$\tau^{\leq n-1} \tau^{\leq n} E \rightarrow \tau^{\leq n} E \rightarrow \tau^{\geq n} \tau^{\leq n} E \rightarrow (\tau^{\leq n-1} \tau^{\leq n} E)[1]$$

and we show $\tau^{\leq n-1} \tau^{\leq n} E = \tau^{\leq n-1} E$, so if we call the terms E^{n-1}, E^n, A^n respectively, we get our desired triangle with $A^n \in \mathcal{A}^\sharp[-n]$. The fact that $\tau^{\leq n-1} \tau^{\leq n} E = \tau^{\leq n-1} E$ is essentially the fact that $i^n i^{n-1} = i^n$, i.e., we argue by Yoneda

$$\begin{aligned} \mathrm{Hom}(A, \tau^{\leq n-1} \tau^{\leq n} E) &\cong \mathrm{Hom}(i^{\leq n-1} A, \tau^{\leq n} E) \\ &\cong \mathrm{Hom}(i^{\leq n} i^{\leq n-1} A, E) \\ &\cong \mathrm{Hom}(i^{\leq n} A, E) \\ &\cong \mathrm{Hom}(A, \tau^{\leq n} E). \end{aligned}$$

Now, the bounded assumption will tell us that only finitely many $A_n \neq 0$, and we use these sorts of distinguished with that fact to get the finite filtration.

Now, conversely, let $\mathcal{A}^\sharp$ be a bounded t-structure as in the second definition, and we show $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ satisfies the first. Essentially by construction, we automatically have (ii). To get (i), we can reformulate our definition of $\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0}$ as

$$\begin{aligned} \mathcal{D}^{\leq 0} &= \langle \mathcal{A}^\sharp[k_1] \mid k_1 \geq 0 \rangle \\ \mathcal{D}^{\geq 1} &= \langle \mathcal{A}^\sharp[k_2] \mid k_2 \leq -1 \rangle. \end{aligned}$$

By assumption, we have for $k_1 \geq 0 > -1 \geq k_2$,

$$\mathrm{Hom}(\mathcal{A}^\sharp[k_1], \mathcal{A}^\sharp[k_2]),$$

which implies $\mathcal{A}^\sharp[k_1] \subseteq {}^\perp \mathcal{A}^\sharp[k_2]$ for all $k_1 \geq 0$, so

$$\mathcal{D}^{\leq 0} = \langle \mathcal{A}^\sharp[k_1] \mid k_1 \geq 0 \rangle \subseteq {}^\perp \mathcal{A}^\sharp[k_2],$$

which implies for all $k_2 \leq -1$,

$$\mathcal{A}^\sharp[k_2] \subseteq (\mathcal{D}^{\leq 0})^\perp,$$

so

$$\mathcal{D}^{\geq 1} = \langle \mathcal{A}^\sharp[k_2] \mid k_2 \leq -1 \rangle \subseteq (\mathcal{D}^{\leq 0})^\perp,$$

so indeed

$$\mathrm{Hom}(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 1}).$$

To get (iii), take a cohomology filtration, re-indexed so that $A^i \in \mathcal{A}^\sharp[-i]$ (allowing negative indices and maybe $A_i = 0$),

$$\begin{array}{ccccccc} 0 = E^{-n} & \longrightarrow & E^{-n+1} & \rightarrow \cdots \rightarrow & E^{-1} & \longrightarrow & E^0 \rightarrow \cdots \rightarrow E^{n-1} \longrightarrow E^n = E \\ & \nwarrow \text{dashed} & \swarrow & & \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} \\ & A^{-n} & & & A^0 & & A^n \end{array}$$

and complete $E^0 \rightarrow E$ to a distinguished triangle

$$E^0 \rightarrow E \rightarrow E/E^0 \rightarrow E^0[1].$$

By construction $E^0 \in \mathcal{D}^{\leq 0}$, and as we have argued before, $E/E^0 \in \mathcal{D}^{\geq 1}$. □

Remark 2.31. Let $\mathcal{D}$ be a triangulated category. Given the heart of a bounded t-structure $\mathcal{A}^\sharp$ which induces $(\tau^{\leq 0}, \tau^{\geq 0})$, we had defined for $E \in \mathcal{D}$

$$H_{\mathcal{P}}^i(E) = \tau^{\leq i} \tau^{\geq i} E \in \mathcal{A}^\sharp[-i].$$

We can formulate this differently. If $E \in \mathcal{D}^{\leq n} \cap \mathcal{D}^{\geq -n}$, then it has t-structure cohomology filtration

$$0 = E^{-n} \longrightarrow E^{-n+1} \rightarrow \cdots \rightarrow E^{-1} \longrightarrow E^0 \rightarrow \cdots \rightarrow E^{n-1} \longrightarrow E^n = E$$

$\nwarrow \quad \swarrow \quad \quad \nwarrow \quad \swarrow \quad \quad \nwarrow \quad \swarrow$
 $A^{-n} \quad \quad \quad A^0 \quad \quad \quad A^n$

by choosing $E^i = \tau^{\leq i} E$, and then $A^i \in \mathcal{A}^\sharp[-i]$. We exactly then have

$$H_{\mathcal{P}}^i(E) = A^i[i].$$

This will tell us that we can read off the t-structure cohomology in the t-structure cohomology filtration given by a bounded t-structure—all cohomologies are 0, except for those appearing as cones in the filtration.

We then denote this t-structure cohomology by

$$H_{\mathcal{A}^\sharp}^i = H_{\mathcal{P}}^i,$$

as we will later be discussing multiple t-structures at the same time.

Example 2.32. standard t-structure on $D(\mathcal{A})$

2.2.3 Hearts are Abelian

Theorem 2.33. $\mathcal{A} = \mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$ is an abelian category.

Example 2.34. To motivate the proof, let's try to use the standard t-structure/truncations to find kernels/cokernels for the standard heart of the derived category.

That is, let $\mathcal{A}$ be an abelian category, $D(\mathcal{A})$ the derived category, and equip $D(\mathcal{A})$ with the standard t-structure $(D(\mathcal{A})^{\leq 0}, D(\mathcal{A})^{\geq 0})$, which has as heart $\mathcal{A}$ (realized as the degree 0 complexes).

Given $f : A \rightarrow B$, we form the cone $C(f)$ as the complex

$$0 \rightarrow A \rightarrow B \rightarrow 0,$$

with A in degree -1 , B in degree 0. Then, we get

$$\begin{aligned} \tau_{\leq -1} C(f) &= (0 \rightarrow \ker f \rightarrow 0 \rightarrow 0) \\ \tau_{\geq 0} C(f) &= (0 \rightarrow 0 \rightarrow \operatorname{coker} f \rightarrow 0), \end{aligned}$$

so we see $(\tau_{\leq -1} C(f))[1]$ (shifted to be a degree 0 complex) recovers the kernel, and $\tau_{\geq 0} C(f)$ recovers the cokernel.

We will see that this works for any t-structure.

Proof. This is an additive category, since it is the intersection of two additive subcategories. So, we just have to show that it admits kernels, cokernels, and a morphism decomposition making image canonically isomorphic to coimage.

Let $f : X \rightarrow Y$ be a morphism in $\mathcal{A}$, denote Z its cone, and define

$$K = \tau^{\leq -1} Z \quad C = \tau_{\geq 0} Z.$$

Define k as the composition $\tau_{\leq -1} Z \rightarrow Z \rightarrow X[1]$, and c as the composition $Y \rightarrow Z \rightarrow \tau_{\geq 0} Z$. We claim that $(K[-1], k[-1])$ and (C, c) are the kernel and cokernel of f .

Let's show (C, c) is the cokernel, the kernel is showed similarly. By construction $C \in \mathcal{D}^{\geq 0}$, and from the triangle

$$Y \rightarrow Z \rightarrow X[1] \rightarrow Y[1]$$

where $Y, X[1] \in \mathcal{D}^{\leq 0}$, we need $Z \in \mathcal{D}^{\leq 0}$. □

2.2.4 Tilting at a Torsion Pair

One large source of examples of t-structures is by what is called “tilting at a torsion pair”.

Definition 2.35. Let $\mathcal{A}$ be an abelian category. A **torsion pair** $(\mathcal{T}, \mathcal{F})$ is a pair of (strictly) full subcategories of $\mathcal{A}$ satisfying

- (i) $\text{Hom}(\mathcal{T}, \mathcal{F}) = 0$
- (ii) Every object $A \in \mathcal{A}$ fits in a short exact sequence

$$0 \rightarrow T \rightarrow A \rightarrow F \rightarrow 0$$

for some $T \in \mathcal{T}$ and $F \in \mathcal{F}$.

Now, we explain how to tilt at a torsion pair.

Theorem/Definition 2.1. Let $\mathcal{D}$ be a triangulated category, and let $\mathcal{A}$ be the heart of a bounded t-structure on $\mathcal{D}$. Let $(\mathcal{T}, \mathcal{F})$ be a torsion pair on $\mathcal{A}$. Then, the following is the heart of a bounded t-structure on $\mathcal{D}$:

$$\mathcal{A}^\sharp = \langle \mathcal{F}[1], \mathcal{T} \rangle = \{X \in \mathcal{D} \mid H_{\mathcal{A}}^0(X) \in \mathcal{T}, H_{\mathcal{A}}^{-1}(X) \in \mathcal{F}, H_{\mathcal{A}}^i(X) = 0 \text{ otherwise}\}.$$

Proof. We give a sketch.

First, $\text{Hom}(\mathcal{A}^\sharp, \mathcal{A}^\sharp[i]) = 0$ for $i < 0$ because

$$\text{Hom}(\mathcal{F}[1], \mathcal{F}[1+i]) = \text{Hom}(\mathcal{F}[1], \mathcal{T}[i]) = \text{Hom}(\mathcal{T}, \mathcal{T}[i]) = \text{Hom}(\mathcal{T}, \mathcal{F}[1+i]) = 0,$$

(where the last equality comes from the definition of a torsion pair) imply

$$\text{Hom}(\langle \mathcal{F}[1], \mathcal{T} \rangle, \langle \mathcal{F}[1], \mathcal{T} \rangle [i]) = 0.$$

Now, we need to produce t-structure cohomology filtrations for $\mathcal{A}^\sharp$. Let $E \in \mathcal{D}$. Our bounded t-structure $\mathcal{A}$ gives us a filtration

$$\begin{array}{ccccccc}
0 = E^{-n} & \longrightarrow & E^{-n+1} & \longrightarrow & \cdots & \longrightarrow & E^{-1} & \longrightarrow & E^0 & \longrightarrow & \cdots & \longrightarrow & E^{n-1} & \longrightarrow & E^n = E \\
& \nwarrow \text{dashed} & & \swarrow & & & \nwarrow \text{dashed} & & \swarrow & & & & \nwarrow \text{dashed} & & \swarrow \\
& & A^{-n} & & & & & A^0 & & & & & & A^n & &
\end{array}$$

where $A^i \in \mathcal{A}[-i]$. Then, we “refine” the filtration using the torsion pair as follows. By the torsion pair, we have a short exact sequence

$$0 \rightarrow T^i \rightarrow A^i \rightarrow F^i \rightarrow 0,$$

which will give us a distinguished triangle in $\mathcal{A}$. Thinking of E^i as an extension of A^i by E^{i-1} , we will lift come up with $\tilde{T}^i$ which is an extension of T^i by E^{i-1} , such that we have distinguished triangles.

$$\begin{array}{ccccc}
E^{i-1} & \longrightarrow & \tilde{T}^i & \longrightarrow & E^i \\
& \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} & \swarrow \\
& & T^i & & F^i
\end{array}$$

We can do this by defining

$$\tilde{T}^i = C(T^i[-1] \rightarrow E^{i-1}),$$

and then getting a map $\tilde{T}^i \rightarrow E^i$ by pulling back $E^{i-1} \rightarrow E^i$, after noticing that the composition $T^i[-1] \rightarrow E^{i-1} \rightarrow \tilde{T}^i$ is 0 (and using that Hom is a cohomological functor). Then, the cone over $E^{i-1} \rightarrow \tilde{T}^i$ is T^i by definition, and the cone over $\tilde{T}^i \rightarrow E^i$ is F^i by the octahedral axiom (i.e., analyze the composition $E^{i-1} \rightarrow \tilde{T}^i \rightarrow E^i$).

The point now is to filter E by the $\tilde{T}^i$ instead of the E^i .

Putting two of these diagrams next to each other and deleting the left/right ends, we get the following.

$$\begin{array}{ccccc}
\tilde{T}^{i-1} & \longrightarrow & E^{i-1} & \longrightarrow & \tilde{T}^i \\
& \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} & \swarrow \\
& & F^{i-1} & & T^i
\end{array}$$

We now need to compute the cone over $\tilde{T}^{i-1} \rightarrow \tilde{T}^i$, which we denote by $\tilde{A}^i$. Applying the octahedral axiom, we get a distinguished triangle

$$F^{i-1} \rightarrow \tilde{A}^i \rightarrow T^i \rightarrow F^{i-1}[1].$$

Taking $\mathcal{A}$ cohomology, and recognizing that

$$H_{\mathcal{A}}^j(F^{i-1}) = \begin{cases} F^{i-1}[i-1] & j = i-1 \\ 0 & \text{otherwise} \end{cases}, \quad H_{\mathcal{A}}^j(T^i) = \begin{cases} T^i[i] & j = i \\ 0 & \text{otherwise} \end{cases}$$

we then get exact sequences

$$0 \rightarrow F^{i-1}[i-1] \rightarrow H_{\mathcal{A}}^{i-1}(\tilde{A}^i) \rightarrow 0$$

and

$$0 \rightarrow H_{\mathcal{A}}^i(\tilde{A}^i) \rightarrow T^i[i] \rightarrow 0,$$

so in other words

$$\begin{aligned} H_{\mathcal{A}}^{-1}(\tilde{A}^i[i]) &= F^{i-1}[i-1] \in \mathcal{F} \\ H_{\mathcal{A}}^0(\tilde{A}^i[i]) &= T^i[i] \in \mathcal{T}, \end{aligned}$$

so indeed

$$\tilde{A}^i \in \mathcal{A}^{\sharp}[-i],$$

so we get an $\mathcal{A}^{\sharp}$ t-structure cohomology filtration as desired. $\square$

Tilting provides tons of examples of t-structures—starting with the standard t-structure $\mathcal{A} \subseteq D^b(\mathcal{A})$, we can tilt over and over again to find more and more t-structures. One question we can ask is whether two given t-structures are tilts of each other. We can use the following proposition to help us answer this questions.

Proposition 2.36. Let $\mathcal{A}^{\sharp}, \mathcal{A}$ be hearts of bounded t-structures on a triangulated category $\mathcal{D}$, and assume

$$\mathcal{A}^{\sharp} \subseteq \langle \mathcal{A}[1], \mathcal{A} \rangle.$$

Then, $\mathcal{A}^{\sharp}$ is a tilt of $\mathcal{A}$ at the torsion pair

$$\mathcal{T} = \mathcal{A} \cap \mathcal{A}^{\sharp}, \quad \mathcal{F} = \mathcal{A} \cap \mathcal{A}^{\sharp}[-1].$$

Proof. It's clear that

$$\mathrm{Hom}(\mathcal{T}, \mathcal{F}) = 0,$$

since $\mathcal{T} \subseteq \mathcal{A}^{\sharp}$ and $\mathcal{F} \subseteq \mathcal{A}^{\sharp}[-1]$.

Now, we need to show that every object $A \in \mathcal{A}$ is an extension of $\mathcal{F}$ by $\mathcal{T}$. To do this, we study the $\mathcal{A}^{\sharp}$ cohomology filtration of A , which will be something like

$$\begin{array}{ccccccc} \cdots & \rightarrow & A^{-2} & \xrightarrow{\quad} & A^{-1} & \xrightarrow{\quad} & A^0 & \xrightarrow{\quad} & A^1 & \xrightarrow{\quad} & A^2 & \rightarrow \cdots \\ & & \nwarrow \text{dashed} & & \swarrow & \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} & \swarrow & \\ & & H^{-1} & & H^0 & & H^1 & & H^2 & & \end{array}$$

and we want to argue that only H^0, H^1 are nonzero, so that we get a distinguished triangle

$$H^0 \rightarrow A \rightarrow H^1 \rightarrow H^0[1],$$

FINISH THIS!! $\square$

2.2.5 Examples in Algebraic Geometry

Example 2.37. Let (X, H) be a polarized surface. We will define a tilt of $D^b(X)$ which will be used later to define a stability condition.

Let $\mu \in (-\infty, \infty]$. Then, define

$$\mathcal{T}_\mu = \{\text{torsion sheaves}\} \cup \{\text{sheaves with } \mu^- > \mu\}$$

3 Stability in General

We will now extend the notion of a stability condition on an abelian category $\mathcal{A}$ to a stability condition on a triangulated category $\mathcal{D}$. The story goes that choose a stability condition on a triangulated category is exactly the same as choosing a t-structure on $\mathcal{D}$ to get an abelian heart $\mathcal{A}^\sharp$, and then choosing a stability condition on $\mathcal{A}^\sharp$ (so we're “refining” the t-structure), though we will give a more symmetric/elegant definition.

Then, our task will be to classify stability conditions. This will be very interesting from the perspective of birational geometry—a given stability condition σ on the derived category of coherent sheaves on a surface $D^b(\text{Coh}(S))$ will give a notion of stable objects, which will then have a fine moduli space. When we nudge the stability condition only a little bit, we will typically have the same stable objects and so the same moduli space. However, if we change it enough we will get different collections of stable objects, so a different moduli space which will turn out to be birational to our previous one. So, to study the birational geometry of these moduli spaces, studying *all* of the stability conditions will be very valuable.

Now, the set of all stability conditions turns out to naturally have the structure of a complex manifold. Proving this will be one of our first tasks.

Definition 3.1. Let $\mathcal{D}$ be a triangulated category. A **slicing** on $\mathcal{D}$ is a collection $\mathcal{P} = \bigcup_{\phi \in \mathbb{R}} \mathcal{P}(\phi)$ of full additive subcategories, satisfying

- (i) $\mathcal{P}(\phi + 1) = \mathcal{P}(\phi)[1]$
- (ii) For $\phi_1 > \phi_2$, $\text{Hom}(\mathcal{P}(\phi_1), \mathcal{P}(\phi_2)) = 0$
- (iii) Harder-Narasimhan filtrations exist, i.e., for all $E \in \mathcal{D}$, there exist $\phi_1 > \phi_2 > \dots > \phi_n$ together with a diagram of triangles

$$\begin{array}{ccccccc} 0 = E^0 & \longrightarrow & E^1 & \longrightarrow & E^2 & \longrightarrow \dots \longrightarrow & E^{n-1} & \longrightarrow & E^n = E \\ & & \swarrow & & \swarrow & & \swarrow & & \swarrow \\ & & A^1 & & A^2 & & A^n & & \end{array}$$

where $A^i \in \mathcal{P}(\phi_i)$.

The nonzero objects of $\mathcal{P}(\phi)$ are called the **semistable objects of phase ϕ** . For $E \in \mathcal{D}$, denote $\phi^+(E)$ the greatest phase, $\phi^-(E)$ the smallest phase.

Denote $\mathcal{P}(a, b]$ the full subcategory of objects whose Harder-Narasimhan filtrations have associated graded pieces with phases $\phi \in (a, b]$. Equivalently, it is the full extension-closed subcategory generated by all $\mathcal{P}(\phi)$ for $\phi \in (a, b]$.

A **central charge** is a group homomorphism $Z : K(\mathcal{D}) \rightarrow \mathbb{C}$, together with the data of a group homomorphism $\lambda : K(\mathcal{D}) \rightarrow \Lambda$ to a finite rank lattice, and $z : \Lambda \rightarrow \mathbb{C}$, where we require $Z = z \circ \lambda$.

A **pre-stability condition** is a pair $(Z, \mathcal{P})$ of a central charge and slicing such that for a nonzero $E \in \mathcal{P}(\theta)$, $Z(E) \in \mathbb{R}_{>0} \cdot e^{i\pi\theta}$.

Remark 3.2. Extra conditions are usually enforced to before calling a pre-stability condition a stability conditions. These conditions will make stability conditions deform well.

Remark 3.3. Harder-Narasimhan filtrations are unique up to isomorphism. The proof is exactly the same as the uniqueness for the bounded t-structure cohomology filtration.

Example 3.4. Let $\mathcal{D}$ be a triangulated category, $\mathcal{A}$ the heart of a bounded t-structure, and $Z_{\mathcal{A}} : K(\mathcal{A}) = K(\mathcal{D}) \rightarrow \mathbb{C}$ a stability condition on the heart. Then, we can define a pre-stability condition (not necessarily satisfying the support property to be defined later) on $\mathcal{D}$ as follows:

- (i) Define the slicing $\mathcal{P}$ by setting, for $\phi \in (0, 1]$, $\mathcal{P}(\theta) := \{Z_{\mathcal{A}}\text{-semistable objects}\}$, and extend this to all real numbers by $\mathcal{P}(\phi + n) := \mathcal{P}(\phi)[n]$.
- (ii) Define $Z = Z_{\mathcal{A}}$ by identifying $K(\mathcal{D}) = K(\mathcal{A})$, since every object in $\mathcal{D}$ has a $\mathcal{A}$ filtration by definition of a bounded t-structure.

Conversely, given a (pre)stability condition, we get the heart of a bounded t-structure by $\mathcal{A} = \mathcal{P}(0, 1]$, and a stability function $Z_{\mathcal{A}}$, since $K(\mathcal{A}) = K(\mathcal{D})$ for bounded t-structures.

In particular, we now have an example of a (pre)stability condition on $D^b(\mathbb{P}^1)$.

Remark 3.5. Denote $\mathrm{GL}_2^+(\mathbb{R})$ the group of orientation preserving linear automorphisms of $\mathbb{R}^2$. In particular, this group acts on $\mathbb{C}^*$ and $S^1 \subseteq \mathbb{C}^*$ by including, acting, and retracting along rays through the origin. Then, $\widetilde{\mathrm{GL}_2^+(\mathbb{R})}$ acts on $\widetilde{\mathbb{C}^*}$ and $\widetilde{S^1} = \mathbb{R}$. This gives an action of $\mathrm{GL}_2^+(\mathbb{R})$ on pre-stability conditions, via

$$\tilde{g} \cdot (Z, \mathcal{P}) = (g \circ Z, \mathcal{P} \circ \tilde{g}),$$

so that if $E \in \mathcal{P}(\tilde{g}\phi)$ nonzero, then $g \circ Z(E) \in \mathbb{R}_{>0} \cdot e^{i\pi\tilde{g}\cdot\phi}$.

3.1 Deformation

We will now enforce a condition on (pre) stability conditions so that we can prove that stability conditions form a complex manifold in a natural way. First, we formulate the definitions so that we can limit our attention to stability conditions, not all pre-stability conditions. Next, we will give the set of stability conditions a topology, and prove our deformation result.

Definition 3.6. We say a pre-stability condition $(Z, \mathcal{P})$ satisfies the **support property** if, for some norm $\|\cdot\|$ on Λ , we have

$$\inf \left\{ \frac{|Z(E)|}{\|[E]\|} \mid E \text{ is semistable} \right\} > 0.$$

A pre-stability condition with the support property is called a **stability condition**.

Definition 3.7. Let $Q : \Lambda_{\mathbb{R}} \rightarrow \mathbb{R}$ be a quadratic form. We say a pre-stability condition $(Z, \mathcal{P})$ satisfies the support property with respect to Q if

- (i) the kernel $\ker z \leq \Lambda_{\mathbb{R}}$ of the central charge is negative definite with respect to Q ,
- (ii) for any semistable object E , we have $Q(E) \geq 0$.

Example 3.8. The prestability condition we put on $\mathbb{P}^1$ clearly has the support property—the infimum $1 > 0$. As we will see in the proposition

Proposition 3.9. A pre-stability condition $(Z, \mathcal{P})$ has the support property iff it has the support property for some quadratic form Q .

Proof. Assume $(Z, \mathcal{P})$ has the support property. So, for semistable objects E , we have

$$\begin{aligned} \frac{|Z(E)|}{\|[E]\|} &\geq c > 0 \\ |Z(E)|^2 &\geq c^2 \|[E]\|^2 \\ |Z(E)|^2 - c^2 \|[E]\|^2 &\geq 0. \end{aligned}$$

Now, define Q by

$$Q(E) = |Z(E)|^2 - c^2 \|[E]\|^2,$$

so by construction it is positive definite on semistable objects. Furthermore, Q is negative definite on $\ker z$ because $|Z(E)| = 0$.

Conversely, assume $(Z, \mathcal{P})$ has the support property for some Q . Then, one can define a norm on Λ by

$$\|[E]\|^2 := |Z(E)|^2 - Q(E).$$

□

There is then a generalized metric (i.e., allowing ∞) on the set of slicings by

$$d_S(\mathcal{P}, \mathcal{Q}) = \sup_{0 \neq E \in \mathcal{D}} \{ |\phi_{\mathcal{Q}}^-(E) - \phi_{\mathcal{P}}^-(E)|, |\phi_{\mathcal{Q}}^+(E) - \phi_{\mathcal{P}}^+(E)| \} \in [0, +\infty].$$

In fact, this is equivalent to

$$d_S(\mathcal{P}, \mathcal{Q}) = \sup_{\phi \in \mathbb{R}} \sup_{\substack{E \neq 0 \\ E \in \mathcal{P}(\phi)}} \{ |\phi_{\mathcal{Q}}^+(E) - \phi|, |\phi_{\mathcal{Q}}^-(E) - \phi| \}.$$

Now that we've equipped the set of slicings, denoted $\text{Slice}(\mathcal{D})$, with a topology, we equip $\text{Stab}_\Lambda(\mathcal{D})$ with the coarsest topology that makes the following maps continuous.

$$\begin{aligned}\text{Stab}_\Lambda(\mathcal{D}) &\rightarrow \text{Slice}(\mathcal{D}), & (Z, \mathcal{P}) &\mapsto \mathcal{P} \\ \mathcal{Z} : \text{Stab}_\Lambda(\mathcal{D}) &\rightarrow \text{Hom}(\Lambda, \mathbb{C}), & (Z, \mathcal{P}) &\mapsto Z.\end{aligned}$$

where $\text{Hom}(\Lambda, \mathbb{C})$ has the operator norm between the normed vector spaces, where Λ has its fixed norm $\|\cdot\|$.

In particular, $\text{Stab}_\Lambda(\mathcal{D})$ is topologized by the following metric. For $\sigma = (Z, \mathcal{P})$ and $\tau = (W, \mathcal{Q})$ stability conditions,

$$d(\sigma, \tau) := \max\{d_S(\mathcal{P}, \mathcal{Q}), \|Z - W\|\}.$$

We will now work towards, as promised, the following result:

Theorem 3.10. Let Q be a quadratic form on $\Lambda_{\mathbb{R}}$. Assume $\sigma = (Z, \mathcal{P})$ satisfies the support property with respect to Q . Then,

- (i) There is an open neighborhood $\sigma \in U_\sigma \subseteq \text{Stab}_\Lambda(\mathcal{D})$ such that the restriction $\mathcal{Z} : U_\sigma \rightarrow \text{Hom}(\Lambda, \mathbb{C})$ is a covering of the set of Z' such that Q is negative definite on $\ker z'$.
- (ii) All stability conditions in U_σ satisfy the support property with respect to Q .

In particular, $\mathcal{Z}$ is a local homeomorphism, and $\text{Stab}_\Lambda(\mathcal{D})$ is a manifold.

This first lemma will help us understand when two stability conditions with nearby central charges are in the same sheet, anticipating that $\mathcal{Z}$ will be a covering map.

Lemma 3.11. Assume that $\sigma = (Z, \mathcal{P})$ and $\tau = (W, \mathcal{Q})$ are two pre-stability conditions such that σ satisfies the support property with respect to Q , such that

$$\frac{|W(v) - Z(v)|}{|Z(v)|} < \sin \pi \varepsilon \quad \text{for all } v \in \Lambda \text{ with } Q(v) \geq 0,$$

and such that either $d(\mathcal{P}, \mathcal{Q}) < \frac{1}{4}$, or that σ, τ has the same heart $\mathcal{P}(0, 1] = \mathcal{Q}(0, 1]$. Then $d(\mathcal{P}, \mathcal{Q}) < \varepsilon$.

As motivated above, we have the following corollary. The yet to be proven covering map is locally injective, and for U “small”, a local lift (if it exists) is continuous.

Corollary 3.12. The map $\mathcal{Z} : \text{Stab}_\Lambda(\mathcal{D}) \rightarrow \text{Hom}(\Lambda, \mathbb{C})$ is locally injective.

Moreover, consider a section $U \rightarrow \text{Stab}_\Lambda(\mathcal{D})$, $Z \mapsto \sigma_Z(Z, \mathcal{P}_Z)$ such that every σ_Z satisfies the support property with respect to Q . Assume that U can be covered by open subsets V_i such that for all $Z, Z' \in V_i$, we have $d(\mathcal{P}_Z, \mathcal{P}_{Z'}) < \frac{1}{4}$. Then, this section is continuous.

4 Stability on Surfaces

5 Stability on Projective Space

5.1 Heart of Linear Complexes

We recall a theorem of Beilinson. First, we define some notation.

- Let V be a vector space (whose elements we consider as geometric vectors), and $V^\vee$ its dual, and set $S = \text{Sym}^\bullet V^\vee$ as usual.
- Let $\text{Gr}S\text{-Mod}$ the category of graded (left) modules over S .
- Let $M_{[0,n]}(S)$ be the full subcategory of $\text{Gr}S\text{-Mod}$ consisting of objects which are finite direct sums of $S(-r)$ for $r \in \{0, \dots, n\}$.
- Let $K_{[0,n]}^b(S)$ the homotopy category of bounded complexes in $M_{[0,n]}(S)$.

We can then define an additive functor

$$\begin{aligned} M_{[0,n]}(S) &\longrightarrow \text{Coh}(\mathbb{P}^n) \\ S(-r) &\longmapsto \widetilde{S(-r)} = \mathcal{O}(-r), \end{aligned}$$

which then passes to chain complex categories and then to homotopy categories (which are triangulated, by source-cylinder-cone-suspension exact triangles)

$$K_{[0,n]}^b(S) \longrightarrow K^b(\text{Coh}(\mathbb{P}^n)),$$

which we can post-compose with the localization functor to get

$$F : K_{[0,n]}^b(S) \longrightarrow D^b(\mathbb{P}^n).$$

In total, this functor just hits a chain complex with usual projective space sheafification/tilde, and likewise for morphisms. The left is relatively easy to understand—morphisms are homotopy classes of honest chain maps—whereas the right is a-priori difficult. The surprising thing is the following.

Theorem 5.1. The functor F is an equivalence.

In other words, every object of $D^b(\mathbb{P}^n)$ can be represented as a complex only with direct sums $\mathcal{O}(-i)$ for $i = 0, \dots, n$, and when we restrict to these representatives, morphisms are much easier to understand—rather than roofs, they are just homotopy classes of chain maps.

Definition 5.2. Define the full subcategory $\mathcal{L} \subseteq D^b(\mathbb{P}^n)$

$$\mathcal{L} = \{0 \rightarrow V^{-n} \otimes_k \mathcal{O}(-n) \rightarrow \dots \rightarrow V^{-1} \otimes_k \mathcal{O}(-1) \rightarrow V^0 \otimes_k \mathcal{O} \rightarrow 0\}.$$

(where V^i are finite dimensional k -vector spaces) called the **category of linear complexes**.

Some things to notice about $\mathcal{L}$. First, all objects in $\mathcal{L}$ are chain complexes whose components are direct sums of $\mathcal{O}(-i)$ for $i = 0, \dots, n$, so by Beilinson, whenever $X, Y \in \mathcal{L}$ (or even in homological degree shifts of $\mathcal{L}$)

$$\mathrm{Hom}_{D^b(\mathbb{P}^n)}(X, Y) = \mathrm{Hom}_{K^b(\mathbb{P}^n)}(X, Y).$$

Even better, there are no nonzero chain homotopies between a given X and Y , since the components of such a chain homotopy would involve morphisms

$$X^{-i} \otimes_k \mathcal{O}(-i) \rightarrow Y^{-i-1} \otimes_k \mathcal{O}(-i-1),$$

of which there is only the 0 map (since there is only the 0 map $\mathcal{O}(-i) \rightarrow \mathcal{O}(-i-1)$). So, in fact,

$$\mathrm{Hom}_{D^b(\mathbb{P}^n)}(X, Y) = \mathrm{Hom}_{C^b(\mathbb{P}^n)}(X, Y),$$

which makes $\mathcal{L}$ truly the category of chain complexes, where there's no need to mod out homotopy or add quasi-inverses.

Even more, we have the following proposition.

Proposition 5.3. $\mathcal{L}$ is the heart of a bounded t-structure.

Proof. First, we show $\mathrm{Hom}_{D^b(\mathbb{P}^n)}(\mathcal{L}, \mathcal{L}[k]) = 0$ for $k < 0$. Let $X, Y \in \mathcal{L}$, and as in the remark

$$\mathrm{Hom}_{D^b(\mathbb{P}^n)}(X, Y[k]) = \mathrm{Hom}_{K^b(\mathbb{P}^n)}(X, Y[k]),$$

and for $k < 0$, the components of our chain maps will be of the form

$$X^{-i} \otimes_k \mathcal{O}(-i) \longrightarrow Y^{-i+k} \otimes_k \mathcal{O}(-i+k)$$

which have to be 0 because there are no morphisms $\mathcal{O}(-i) \rightarrow \mathcal{O}(-i+k)$ for $k < 0$.

Next, we show that every $E \in D^b(\mathbb{P}^n)$ has a t-structure cohomology filtration with respect to $\mathcal{L}$. By Beilinson, every E has a quasi-isomorphism representative where each term is in $M_{[0,n]}(S)$. In the category of complexes, we have a filtration (i.e., actual inclusions) of E where the filtered pieces are the linear strands, which are in $\mathcal{L}$. $\square$

5.2 Stability Condition

Now that we have a heart, we can write down stability conditions. Recall that

$$\mathrm{ch} : K(\mathrm{Coh}(\mathbb{P}^n)) \rightarrow H^*(\mathbb{P}^n; \mathbb{Q})$$

is an isomorphism onto its image. We claim that $\mathcal{O}(-n), \dots, \mathcal{O}$ is a basis for $K(\mathrm{Coh}(\mathbb{P}^n))$. To see this we can check that they have linearly independent chern characters,

$$K(\mathrm{Coh}(\mathbb{P}^n)) = \langle \rangle$$